

AUTOMATIC TRASHCAN SYSTEM

A Project Proposal Presented to
the Faculty of the Integrated Programming and Technologies 1
PAMPANGA STATE UNIVERSITY
STO. TOMAS CAMPUS
Sto. Tomas, Pampanga

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INTRODUCTION

The world is continuously being globalized and this movement is tied with fast-paced developments and innovations that concern both the people and each nation. This change also roots different global issues and concerns and one of these is the problem with Solid Waste Management. The population is perpetually increasing both in the urban and rural areas. As indicated by Khan et al. (2022), the world is rapidly urbanizing. Cities are now under pressure to properly manage trash on a social and environmental level as a result of these developments. This does not just concern every individual but also the different institutions and groups responsible in the community. According to Dolipas et al. (2020), waste management is a crucial issue for any institution or society. It is conceivable that waste management might function to maintain cleanliness and order in a community or institution, conserve resources by employing recyclable materials, or protect the environment in response to the global climate change that is now occurring. (Pardini and others, 2019). The Philippines garbage output is steadily increasing and is predicted to do so in the years to come, according to Coracero et al. (2021). According to the report, problems with solid waste management include the nation's increasing volume of solid waste, inadequate legal enforcement, a shortage of sanitary landfills, and improper disposal. The Ecological Solid trash Management Act of 2000, also known as RA 9003, is the nation's best option and emphasises the need of trash reduction, appropriate disposal, and segregation. Waste collection, waste segregation, and waste disposal are all steps in the proper waste management process. According to Dolipas et al. According to

2019), segregation is the process of separating different components present in solid waste at the site of origin in order to reduce the quantity of garbage that has to be collected and disposed of and to promote recycling and resource reuse. According to Sampulna (2022), interim secretary of the Department of Environment and Natural Disaster (DENR), solid waste management remains a significant issue in the Philippines, mostly because of insufficient municipal trash segregation management. In order to properly address and enforce steps that would ease the Solid Waste Management issue, the institution has encountered challenges. Having effective and efficient solid waste management is one of the first things these organizations can do to maintain the campus clean and free of any harmful substances that may jeopardize the health of its people. According to Molina et al. (2021), Republic Act No. 9003 section 55 requires the Department of Education (DepEd), Commission on Higher Education (CHED), and other national organizations to put in place programs for information exchange and ongoing education on solid waste management. Rey De Jesus, the head of GIMU, stated that the most commonly disposed of types of waste each day include pet bottles, plastic cups, styrofoam, paper, boxes, and tree leaves. Waste is collected daily from Monday to Sunday, between 9:00 AM and 4:00-5:00 PM. However, the proper implementation of RA Act No. 9003 has faced challenges due to: lack of awareness, even though it is covered in the curriculum, students might not understand how important it is to dispose of waste properly; large numbers of students, ensuring that every student understands and complies with waste segregation procedures can be challenging, attitudes and behavioral issues, waste segregation may be inconvenient for certain employees and

students, or they may not view it as their duty; large amount of waste, due to everyday operations schools. frequently produce large volumes of waste, which makes management difficult. This lowers the efficiency of segregation initiatives and results in inappropriate disposal practices.

The researchers aimed to innovate the traditional waste management of the campuses using the technology, by developing a automatic trashcan system. This would not abandon traditional waste management; instead, it would be put into innovation to come up with a new organized proper waste handling that will take action from collection of waste to segregation of what is plastic, metal, paper, and others. Traditional waste management faces inefficiencies, primarily due to static collection schedules and a lack of real-time monitoring on fill levels, leading to overflowing trash bins. To address these challenges, a automatic trashcan system offers an innovative solution by integrating a automatic trashcan that is powered by an independent sustainable energy source, deploying sensors in trash bins, real- time monitoring of fill levels, and implementing automatic sorting technologies that will improve waste segregation and reduce contamination. With that being said, the study aimed to evaluate the effectiveness of the system in improving waste segregation and collection efficiency, providing insights into the impact of technological interventions on sustainable waste management practices.

Purpose and Description

The study aimed to transform waste management practices in PSU Sto.Tomas Campus by utilizing modern technology. The

researchers planned to introduce innovative solutions, including sensor-equipped automatic trash bins, to enhance waste segregation, collection, and disposal processes. The PSU Sto.Tomas waste management staff will be notified when the trash cans reach the maximum fill% thanks to the built-in capacity sensors and automated SMS alerting systems in DSS cans. Furthermore, the garbage cans have the ability to separate various waste materials, including paper, metal, and plastic items. The study also included a Decision Support System (DSS) that, in the event that a sensor in the Automatic Trashcan System failed, instantly notified the PSU Sto.Tomas waste management employees. In order to power these technical upgrades of the equipment, including conveyors and sensors, the researchers included a sustainable energy source. lowering the environmental effect and dependence on traditional power sources Additionally, the research aimed to create a module for tracking the proportion of trash cans that are filled in real time and assess the system's efficacy using ISO 14001 requirements for interaction and usability. By collaborating with PSU Sto.Tomas Campus waste management guidelines, raising awareness among waste disposers, and implementing sustainable energy solutions, the researchers sought to optimize waste management practices and promote environmental sustainability within the campus.

This study provides numerous benefits. First, the use of sensor-equipped trash bins enables timely collection of waste, which minimizes overflow, lowers unpleasant smells, and keeps the campus clean. Second, the segregation function promotes more efficient recycling by reducing the amount of waste sent to

landfills and increasing resource recovery. Third, the installation of sustainable energy sources serves to reduce the cost of energy and the campus's environmental impact, which aligns with sustainable energy efforts. The DSS improves decision-making by giving real-time notifications, allowing waste management workers to respond quickly to problems and maintain uninterrupted operations. Finally, by following ISO 14001 standards, the system not only enhances operational efficiency but also guarantees that waste management methods comply with international environmental standards, establishing a culture of sustainability throughout the campus community.

Here are the user types of the Automatic Trashcan System:

1. Students, Faculty, and Staff

- Is one of the waste depositors.
- This allows the students, faculty, and staff to become more mindful about the way they dispose of their waste.
- The system helps maintain a hygienic campus by efficiently segregating waste, reducing litter, and promoting cleanliness, which improves the overall learning atmosphere.
- By utilizing data, faculty, and staff will have access to valuable insights that can enhance campus policies and operations.

2. Management Personnel

- This allows the utility officer to be in charge of ensuring that reporting and waste management regulations are observed.

- The automated system optimizes the waste segregation process, allowing utility personnel to focus on more important tasks instead of manually sorting waste. This leads to faster and more efficient waste collection and improves overall workflow.
- With the automatic sorting of waste, utility personnel are less likely to handle hazardous or unsanitary materials. This reduces the risk of accidents and health issues, ensuring a safer and healthier work environment.

Objectives of the study

The study's general objective was to improve waste management effectiveness through the implementation of a Smart Waste Management System. The specific objectives of this study were:

1. to align the PSU waste management guidelines with a focus on awareness of waste segregation and proper disposal among students, faculty, and staff;
2. to design and develop the architecture of the prototype, focusing on the effective integration of the hardware components and system functionalities;
3. to develop a module for automated SMS alerts and address issues like busted sensors and overflowing trash bins;
4. to integrate an independent sustainable energy source;
5. to evaluate the system's effectiveness based on ISO 14001 standards for usability and interaction of the prototype.

Significance of the Study

Efficient waste management practices can significantly reduce a campus's environmental impact by minimizing waste that will be sent to landfills, thus preventing soil and water contamination. This study serves as a practical example for students to learn about sustainability and environmental responsibility. Analyzing waste patterns helps identify opportunities for waste reduction and recycling, leading to cost savings and better resource utilization. Sticking to local and national environmental regulations ensures compliance.

This study offers the students practical knowledge in environmental science, sustainability, and waste management, bridging theoretical learning with real-life scenarios. Recycling and trash reduction are examples of eco-friendly practices that are encouraged when people are aware of the negative effects of inappropriate garbage disposal. It also instills a feeling of civic responsibility, inspiring pupils to take an active role in improving the environmental health of their town. As a standard, this study helps assess the effectiveness of current waste management can inspire further researchers into promoting responsible waste disposal behaviors among students and staff. By highlighting current challenges, it encourages innovation in waste management solutions tailored to university environments. Moreover, it adds to the larger conversation on sustainability in higher education by highlighting the part universities play in environmental oversight and establishing guidelines for upcoming studies in this field. The Automatic Trashcan System has a lot to

offer the economy; by enhancing garbage collection routes, reducing fuel consumption, and lowering staff costs, it may lead to more economical operations for waste management companies. Therefore, the Automatic Trashcan System lowers expenses, boosts productivity, and encourages environmental responsibility to build a more resilient and sustainable economy.

The innovative approach to addressing the pressing global issue of solid waste management, particularly in the urban areas where population growth intensifies the challenge. The significance of the study includes:

1. Technological innovation that uses sensor-based sorting systems, renewable energy sources, and automatic technology to construct a automatic trashcan system. This creative method can increase garbage collection efficiency and improve waste segregation, which will improve resource use and lessen the impact on the environment.

2. By using solar panels to implement a sustainable energy solution that supports international efforts to adopt renewable energy, this creative strategy aids in environmental preservation.

3. By adding real-time monitoring systems to trash cans, automatic trashcan can be improved by assuring timely waste collection, lowering the chance of overflow, and minimizing environmental threats. Furthermore, the bins' integrated sorting systems enhance trash segregation, promote proper disposal techniques, and reduce contamination.

4. By reducing the likelihood of contamination and exposure to hazardous materials, efficient waste management contributes to the protection of staff, faculty, and students' health and safety. A safer campus can result from proper waste management, which can help lower the risk of illnesses and pests that are frequently connected to improper trash disposal.

5. A cleaner and more aesthetically pleasing campus environment is the outcome of improved waste management techniques. Sensor-equipped automatic trash cans provide timely waste collection, keep bins from overflowing, and maintain campus cleanliness.

Definition of Terms

1. Compactor - The device starts a condition check when placement is verified. The mechanism that pushes the rubbish into the trash can is activated if the condition is satisfied.

2. Approximate Sensor - To ascertain the garbage cans' capacity, the system monitors their fill levels. It monitors the distance between the garbage within the bin and its top using sensors.

3. Sanitary landfills - These are constructed structures intended for the secure disposal of solid waste; they usually include techniques for waste management and containment to avoid contaminating the environment.

4. Segregation - It is the process of separating various solid waste products at the place of origin in order to make

recycling easier and lower the amount of garbage that has to be gathered and disposed away.

5. RA 9003 or the Ecological Solid Waste Management Act of 2000 – also known as RA 9003, was created to institutionalize and implement sustainable waste management practices throughout the country.

6. Environmental management systems (ISO 14001) - It is designed to enhance how people and systems interact. It focuses on creating user-friendly, effective, and pleasant systems. These guidelines can help researchers make sure their designs are easy to use and encourage a satisfying experience.

7. High Conductance Ultrasonic Sensor (HC-SR04) and JSN-SR04T - These are sensors that use sound waves to determine an object's distance. Whereas the JSN-SR04T functions outside due to its waterproof nature, the HC-SR04 is utilized indoors. Both are often used in distance-measuring tasks.

8. Super Speed Titanium Gear Digital Servo (Savox SC-1258TG)-This is a motor that moves parts in remote-controlled toys like vehicles and airplanes. It is resilient to harsh environments and smooth. It lasts longer without breaking thanks to the titanium gears.

9. ESP32 Camera - The ESP32 Camera module has a built-in camera and Wi-Fi connectivity. In order to classify and separate garbage, this module takes pictures and identifies various waste categories. It is perfect for intelligent waste management applications because of its

effective image processing capabilities and compatibility with devices like the Arduino Mega.

10. Waste Management - The process of gathering, moving, recycling, and getting rid of waste items in an ecologically appropriate manner is known as waste management. It seeks to minimize waste, lessen its negative effects on the environment, and encourage sustainability.

11. Smart Waste Management - It is the process of maximizing trash collection, sorting, and disposal by integrating cutting-edge technology like sensors, the Internet of Things, and data analytics. Through data-driven decision-making and real-time monitoring, it guarantees more sustainable waste management methods, increases efficiency, and lowers operating costs.

Scope and Limitations.

The Automatic Trashcan System utilizes four types of bins to segregate different kinds of waste effectively. The first bin is designated for plastic objects, such as plastic bottles and disposable plastic cups. The second bin is for metal objects, including aluminum cans and steel food cans. The third bin is reserved for paper waste, including paper and other similar materials. Lastly, the fourth bin is for any waste that the sensors cannot detect or classify. The system utilizes solar panels for battery charging, ensuring system maintenance and reducing reliance on external power sources, which supports sustainable operation. Waste is deposited into the machine through a designated opening mechanism, where sensors process it to determine

its composition. The implementation includes HC-SR04 and JSN-SR04T inductive proximity sensors for material identification, distinguishing between plastics, metals, and paper. Additionally, the system employs an ESP32-CAM camera to assist in the identification and classification of waste, especially for more complex materials or when visual identification is necessary. The inclusion of SMS notifications is a critical component, alerting waste management personnel when bins reach 80% capacity to ensure timely collection. The system also incorporates proper waste disposal guidelines to promote responsible usage and support effective waste management practices. A microcontroller module is used for data processing and control, managing the sorting process. The Savox SC-1258TG servo motor drives the conveyor, accelerating waste movement within the machine and ensuring efficient control. The system sends SMS notifications when any bin reaches full capacity, indicating it is full and ready for collection. Additionally, the system has a light indicator that is green and red. When the lights turns green, trash can still be thrown away, but when it turns red, it indicates that the trash bin is full and no longer accepts waste. The Automatic Trashcan System will be utilized by students, faculty, and other individuals at PSU Sto.Tomas Campus, specifically in the College of Computer Studies, to enhance waste management practices and contribute to a cleaner environment.

The system has several limitations; The small size of the trash bin entrance may pose challenges when disposing of oversized or irregularly shaped waste items, as it is inadequate to accommodate certain types of waste effectively; The sensors may not detect certain items, such

as plastic bottles with contents or animals that might enter through the openings, potentially causing alerts or errors: The solar panels may not generate sufficient energy in areas with limited sunlight, such as during rainy days, The lack of available power outlets at the site may lead to downtime or interruptions in system operations if the solar-generated energy is insufficient, The advanced system and various sensors may encounter technical issues such as sensor malfunctions or errors in waste segregation; The conveyor system might experience issues during high waste volumes, leading to delays or congestion in the sorting process; The notification system that sends SMS alerts delay or failures in message delivery, especially with signal or network issues there is also a risk that personnel may not promptly read or act upon notifications, potentially resulting in waste overflow, Furthermore, when the light indicator turns red the system will no longer allow trash to be thrown away, limiting waste disposal when the bins are full.

Users are also advised to follow the recommended guidelines for proper disposal of different types of waste.

Conceptual Framework

The main elements and procedures involved in creating a automatic trashcan system are described in the conceptual framework. The goal of this system was to use modern technology to automate garbage sorting and enhance waste management. Programming, circuit design, and hardware needs, such as sensors and microcontrollers, were all covered in the input. In-depth planning, issue identification, solution analysis, and methodical development including design, coding, testing, and

deployment were all part of the process. The end result of this framework is a fully operational system that effectively classifies garbage and tracks fill levels, guaranteeing that waste management staff receive timely signals for optimal operations.

Tools to Use for the Development of the System

The input included the knowledge requirements in programming languages, circuit design for the schematics, and Arduino microcontroller for the necessary functions. The software requirements were the Arduino IDE. The hardware requirements included HC-SR04 Ultrasonic Sensors, an esp32 camera, proximity sensors for metal, plastic, and paper, an Arduino Uno board, and a Savox SC-1258TG servo motor.

The process involved project planning, identifying the problem and the best solutions, and system development, which included planning and requirements, design, coding, testing, and deployment.

The final output is the Automatic Trashcan System, which sorts waste streams into categories such as plastic, paper, metal, and others, provides the monitoring of data on fill levels and waste composition, and sends notifications to waste collection personnel when bins are full.